Entropy of the Gaussian

I derive the entropy for the univariate and multivariate Gaussian distributions.

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The *information entropy* or *entropy* of a random variable is the average amount information or "surprise" due to the range values it can take. High-probability events have low entropy (not surprising), and low-probability events have high entropy (surprising). For example, if I tell you the sun will rise tomorrow morning, you will be less surprised than if I tell you it will rain. Entropy quantifies your surprise.

The equation for entropy is

$$H(x) = -\int p(x) \log p(x) \mathrm{d}x. \tag{1}$$

Here, the negative sign means high probability events have less entropy. The logarithm ensures that independent events have additive information entropy. If $h(x)$ denotes the information entropy of a single event, then

$$\begin{aligned} h(x, y) &= -\log p(x, y) \\ &= -\log p(x) - \log p(y) \\ &= h(x) + h(y). \end{aligned} \tag{2}$$

The units of entropy depend on the logarithm's base. If $\log = \log_2$, then the units are "bits". If $\log = \log_e = \ln$, then the units are "nats" or *natural units of information*. Let's work through the entropy calculations for the univariate and multivariate Gaussians.

Entropy of the univariate Gaussian. Let x be a Gaussian distributed random variable:

$$x \sim \mathcal{N}(\mu, \sigma^2). \tag{3}$$

Then its entropy is

$$
\begin{aligned}
H(x) &= -\int p(x) \log p(x) \mathrm{d}x \\
&= -\mathbb{E}[\log \mathcal{N}(\mu, \sigma^2)] \\
&= -\mathbb{E}\Big[\log \Big[(2\pi\sigma^2)^{-1/2} \exp \Big(-\frac{1}{2\sigma^2}(x-\mu)^2 \Big) \Big] \Big] \\
&= \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \mathbb{E}[(x-\mu)^2] \\
&\stackrel{\star}{=} \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2}.
\end{aligned} \tag{4}
$$

Step $\star$ holds because $\mathbb{E}[(x-\mu)^2] = \sigma^2$. In words, the entropy of x is just a function of its variance σ^2. This makes sense. As σ^2 gets larger, the range of possible values x can take gets bigger, and the entropy or average amount of surprise increases.

Entropy of the multivariate Gaussian. Now let $\mathbf{x}$ be multivariate Gaussian distributed,

$$
\mathbf{x} \sim \mathcal{N}_D(\boldsymbol{\mu}, \boldsymbol{\Sigma}). \tag{5}
$$

The derivation is nearly the same:

$$
\begin{aligned}
H(\mathbf{x}) &= -\int p(\mathbf{x}) \log p(\mathbf{x}) \mathrm{d}\mathbf{x} \\
&= -\mathbb{E}[\log \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})] \\
&= -\mathbb{E}\Big[\log \Big[(2\pi)^{-D/2} |\boldsymbol{\Sigma}|^{-1/2} \exp \Big(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu}) \Big) \Big] \Big] \\
&= \frac{D}{2} \log(2\pi) + \frac{1}{2} \log |\boldsymbol{\Sigma}| + \frac{1}{2} \mathbb{E}[(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu})] \\
&\stackrel{\star}{=} \frac{D}{2}(1 + \log(2\pi)) + \frac{1}{2} \log |\boldsymbol{\Sigma}|.
\end{aligned} \tag{5}
$$

Step $\star$ is a little trickier. It relies on several properties of the trace operator:

$$
\begin{aligned}
\mathbb{E}[(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu})] &= \mathbb{E}[\operatorname{tr}((\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu}))] \\
&= \mathbb{E}[\operatorname{tr}(\boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^\top)] \\
&= \operatorname{tr}(\boldsymbol{\Sigma}^{-1} \mathbb{E}[(\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^\top]) \\
&= \operatorname{tr}(\boldsymbol{\Sigma}^{-1} \boldsymbol{\Sigma}) \\
&= \operatorname{tr}(\mathbf{I}_D) \\
&= D.
\end{aligned} \tag{6}
$$

Again, we use the fact that $\mathbb{E}[(\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^\top]) = \boldsymbol{\Sigma}$, and again, the average amount of surprise encoded in $\mathbf{x}$ is a function of the covariance.

Notice that entropy is not just a function of $\boldsymbol{\Sigma}$ but the determinant of $\boldsymbol{\Sigma}$. A geometric intuition for the determinant of a covariance matrix is worth its own blog post, but note that $|\boldsymbol{\Sigma}|$ is actually called the *generalized variance*, a scalar which generalizes variance to multivariate distributions (Wilks, 1932). Intuitively, $|\boldsymbol{\Sigma}|$ is analogous to σ^2. As $|\boldsymbol{\Sigma}|$ gets larger, the entropy increases.

1. Wilks, S. S. (1932). Certain generalizations in the analysis of variance. *Biometrika*, 471–494.